

Selective Encryption of H.264/AVC Based on Block Weight Model

Mengdie Huang

School of Information Engineering
Communication University of China
Beijing, China
e-mail: maggie_hmd@163.com

Cheng Yang

School of Information Engineering
Communication University of China
Beijing, China
e-mail: cafeeyang@163.com

Yuan Zhang

School of Information Engineering
Communication University of China
Beijing, China
e-mail: yzhang@cuc.edu.cn

Abstract—The rise of 4K and 8K techniques has led to the growth of video data streaming. Consequently, the greater challenges of encryption efficiency and information leakage facing selective encryption (SE) makes it necessary to reduce the encryption ratio as much as possible. In this paper, we design a SE scheme for H.264/AVC video which achieves a trade-off between low encryption ratio and high safety, for both cryptographic attack and sketch attack point of view. As a starting point, we propose a novel calculation model, called block weight model that can take advantage of reference structure of a video sequence to quantitatively analyze the impact of each block on the motion compensation of other blocks. This calculation model can conduce to accurately select the key bytes in the compressed bitstream for encryption, which greatly retains the network friendliness and improves the efficiency compared to previous work. In addition, our scheme supports scalable encryption granularity by providing an adjustable encryption parameter to meet different security levels. Experimental results verified that our solution is highly efficient in protecting confidential video content with simple computation.

Keywords—selective encryption; H.264/AVC; compressed bitstream; sketch attack

I. INTRODUCTION

According to the latest "Visual Networking Index" report released by Cisco in 2017, mobile video will generate more than three-quarters of data traffic by 2021. What comes with rich video services is the challenging research issue in video security. H.264/AVC becomes common video coding standard thanks to the prominent performance in compression ratios and network friendliness. SE has been proposed and practiced for many decades to protect videos by combining cryptography and video coding techniques. Compared with completely encrypting a bitstream, the strength of SE is the reduction in computational cost of encryption/decryption process.

Currently, SE techniques for H.264/AVC can be categorized into three classes: encryption before compression,

encryption during compression and encryption on compressed bitstream. Schemes that incorporate encryption into coding process usually disturb the statistical properties of syntax elements, resulting in degrading compression ratio. Although Asghar [1] investigated an encryption method that has no effects on the syntax elements statistics, the scheme doesn't favor regular entropy coding engine based on context model. In [2], both regular mode and bypass mode of entropy coding are considered in SE scheme, but the author pointed out that the bitrate indeed increased. What's more, the risk of information leakage resulted from encryption during coding increases as well. The precise choice of syntax elements enables an excellent encryption ratio for encryption in encoding, but syntax elements preserved for meeting format compatibility also give adversaries a chance. The latest sketch attack for H.264 video is presented in [3], which can generate the outline of videos by analyzing the macroblock header. Here, we apply a SE algorithm to the compressed-domain data. It doesn't have any influence on coding efficiency and compression ratio.

The design of selection strategy is based on the analysis of reference structure among the frames in a group of pictures (GOP). In H.264, the reconstruction process of inter frames relies on the motion compensation. Some reconstructed blocks will be frequently referenced by the subsequent blocks. Hence, errors produced in the reconstruction will be accumulated along the path, which is consisted of the subsequent blocks. In contrast, some blocks will be rarely referenced, so the errors will not affect other blocks too much. Obviously, the target of selection encryption should be concentrated on the blocks that are most often referenced.

In this paper, we propose a secure SE scheme for H.264/AVC. Firstly, we present a model for calculating the weight of each block in reconstructing a GOP and then extend the model to calculate the macroblock weight. Secondly, a small number of macroblocks are picked out for encryption with AES according to the weight value. Specially, we provide an encryption parameter to custom the

security level. The remainder of this paper is organized as follows. Section II briefly overviews the prior researches on SE methods for H.264 bitstream. Section III details the proposed block weight model and SE scheme. Section IV describes the experimental results on video appearances after the proposed SE with a detailed analysis of security and computational overhead. Finally, conclusions are drawn in Section V.

II. RELATED WORK

A. Prior Researches on H.264/AVC Video Security

There has been some related work on selectively encrypting codewords of H.264 compression bitstreams. Andreas [4] proposed a SE method for H.264/AVC by changing the NUT value in NAL header and encrypting the NAL unit payload. The approach preserves the network-friendly features of H.264 by encrypting each NAL unit independently. Nevertheless, the encrypt ratio is high due to the lack of selection for payload data. Authors in [5] devised an encryption scheme adaptive to DRM. The granularity of SE can be controlled by customizing parameters of the algorithm. However, in terms of encryption efficiency, the selection strategy based on the slice layer is a little rough, and for every video the encryption location is fixed, without statistical analysis on video content. The idea of selecting Intra Prediction Mode (IPM) to encrypt after CAVLC was applied by Jiang [6]. It can maintain video data format, but the length of codeword will be increased. In addition, all predictive macroblocks (P-MB) and bi-predictive macroblocks (B-MB) are short of protection as the syntax element IPM only exists in the intra macroblock (I-MB). Obviously, the scheme is not sufficient to guarantee the security of video content. The similar problem also exists in [7], encrypting all bits in I-frame with AES. Hole [8] suggested to encrypt entire frame data in I-frame and partial P-frames extracted from H.264 bitstream. It performs well in robustness but takes too much compute overhead. Wang [9] investigated a SE algorithm on Block layer of H.264 to provide multiple levels of video security by shuffling the position of 4x4 Blocks in each MB, but the highest encryption ratio is up to 60% with compression rate dropped by 2%.

Considering that a MB which is decoded from an encrypted MB should be heavily distorted, the works in [10] describes a decision approach in which Dubois encrypting partial MBs for each frame encoded in CAVLC mode, to extend this error through each frame of a video sequence. MB is selected according to the confidentiality which is measured by calculating PSNR. In this method, both encoder frame and decoder need to be redesigned because of the need for one more prediction. Encrypting motion vectors signs was proposed by Khelif [11], who thought that the error from encrypting MVs of the first P frame can be spread. However, when all MVs in P-frame are selected, the encryption ratio is too high, almost 50%. The research [12] shows a motion reference structure for H.264 videos. A ratio named MRR (motion reference ratio) was defined for measuring the dependence between MBs, which is used to select MBs. For

the video with gentle motion, only I-frame will be encrypted. For the videos with intensive motion, all MBs of I-frame and partial MBs of P-frame will be encrypted. If video is encrypted in this way, the motion track will be leaked when only the I frame is selected. Here, the number of selected MBs in I-frame and P-frame also can be decreased by further analysis on MBs.

B. Problems in Selection Encryption of H.264/AVC

We conclude following problems in SE technique for H.264. With more and more Internet traffic being encrypted to protect users' privacy, attacks on encrypted media streams are getting worse. Except conventional cryptographic attacks, some novel perceptual attacks sprung up in recent years [13]. Needless, security must be the first problem to be considered in video SE. What's more, it is the NAL unit that contributes to the adaptability of H.264/AVC bitstreams to heterogeneous networks. Therefore, the encapsulation format of the NAL unit also should be preserved so that it can be properly accepted by the decoder correctly. Computational overhead is a generic problem in designing a scheme and the importance of it will be enhanced in video services with the requirement of low delay. Most of the computational overhead is spent on encryption rather than selection, so the encryption ration is crucial. Usually, a lower encryption ratio can be obtained when selectively encrypt transform coefficients, intra coding modes, or motion vector differences in compressing. But it is also accompanied by high complexity of implementation and impact on compression ratio. Since the issues in SE of H.264/AVC are not independent of each other, we took them into account comprehensively.

III. PROPOSED SELECTIVE ENCRYPTION SCHEM

A. The Model for Calculating the Weight of Blocks

In the H.264 decoder, the distortion of current reconstructed block will be extended to the subsequent blocks which need the current block for reference in reconstruction. In our proposed scheme, we make the most of this feature to scramble the content of video. The Block Weight (BW) was defined to intuitively describe the importance of each block for inter prediction in a group of pictures (GOP). When we reconstruct a 4x4 block, the matching area in the reference frame may correspond to an entire block, or two or four blocks (see Fig. 1).

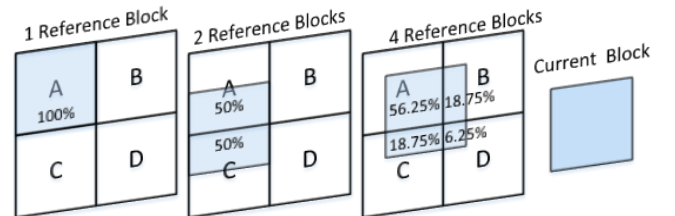


Figure 1. Reference status of reconstructing the block.

Obviously, the proportion of each reference block on reconstructing the current block is not always equal. When the size of covered area in referent block is defined as the

measure, block A plays a more important role in the last reference status because it occupies the largest proportion among four blocks. Additionally, the number of times for each block be referenced is not the same. Some blocks may be used frequently as reference blocks, and some may seldom be referenced. With a view to protect video content by encrypting as few bits as possible, we devised the following model to pick out the blocks with high BW in the GOP.

$$P_{(x,i)} = S_{(x,i)} / 16 \quad (1)$$

$$W_b(x) = \begin{cases} 1 + \sum_{i=1}^n P_{(x,i)} W_{bi}, & n \neq 0 \\ 1, & n = 0 \end{cases} \quad (2)$$

The proportion of Block x referenced by Block i during motion compensation is defined as (1). $S_{(x,i)}$ represents the area in Block x referenced by Block i, the range of which is from zero to sixteen in pixels. When a new Instantaneous Decoding Refresh (IDR) frame of another GOP arrives, the correlation of inter prediction in current GOP will be cut off, so, the BW of Block x is defined within a GOP range as (2). n is the number of blocks which referencing Block x, W_{bi} is the BW of Block i. Since the weight accumulation on the reference block is determined by the motion compensation of the reconstructed blocks, the input to the model should start from the last frame. For keeping the validity of BW after multiplying by the proportion, all of blocks' BW are initialized to 1 instead of 0. Fig.2 presents the process of calculating the BW of each block in a GOP based on this calculation model.

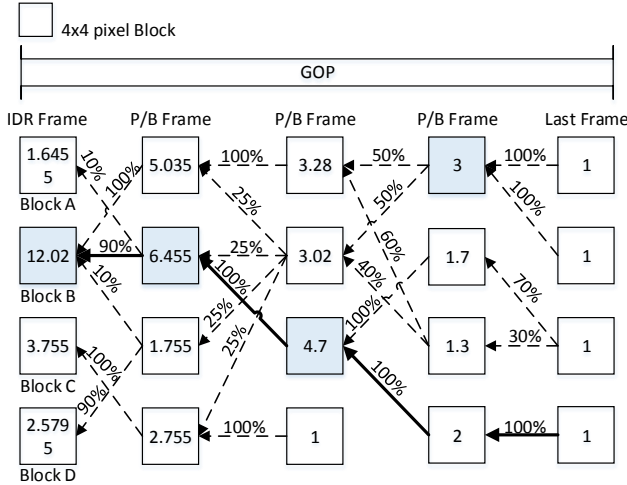


Figure 2. Block weight calculation process.

According to the calculated results, we can draw the conclusion the block with more subsequent blocks and larger referenced proportion, the higher BW of it will be. Meanwhile, depending on the highest BW in the IDR frame, an inter prediction path from the IDR frame to the last frame can be obtained from the model. It means that the distortion

of block B will be extended to the last frame, and to some degree, it has the most significant impact on the prediction of the GOP compared with other three blocks in the same frame. Of course, not every block on the path has the largest BW in its corresponding frame. As the movement changes, some blocks in inter frames will get a higher BW in a short path. For avoid information leakage, it is obviously not enough to encrypt only the blocks in the I-frame. Hence, a secure SE scheme for H.264 video based on the BW model was subsequently proposed by us.

B. Selection Strategy and Encryption Rules

In response to more and more perceived attacks like the sketch attack presented in [3], which relies on the information staying intact in the MB data before and after video encryption, mostly in MB header, we decided to take the MB layer as the granularity of video encryption. Because block layer encryption will inevitably expose MB headers, allowing more hackers to take advantage of it. The value of BW is further processed after the calculation based on the BW model. We defined the MB weight (MBW) of MB x as (3). $W_{b(x,i)}$ is the BW of Block i belonging to MB x. Since the skip blocks will not be encoded in H.264, the BW of Skip Block is defined not being updated in the BW model, keeping the initial value. Based on this initial setting, the motion trace of the objects in the GOP can be observed clearly by calculating the BW of Non-Skip MBs (Non-Skip P-MBs and I-MBs).

$$W_{mb}(x) = (\sum_{i=1}^{16} W_{b(x,i)}) / 16 \quad (3)$$

$$m_n = \alpha N_n = \alpha \cdot (wh / (16 \times 16)), 0 < \alpha \leq 1 \quad (4)$$

$$m_n = \sum_{i=1}^q l_i \quad (5)$$

$$P_s = \sum_{i=0}^{l_s-1} MB_{(s,i)}, s \text{ from } 1 \text{ to } q \quad (6)$$

$$C_s = Enc(P_s), s \text{ from } 1 \text{ to } q \quad (7)$$

$$F_n = \sum_{i=0}^{q-1} C_i \quad (8)$$

After obtaining the reference weights of each MB in a GOP, the decision and encryption steps are as follows.

Step 1 Reorganize the macroblocks (MBs) of each frame into an ordered sequence $\{MB_0, MB_1, \dots, MB_{N_n-1}\}$ through sorting MBs by MBW from high to the low. N_n is the sum of Non-Skip MBs belonging to frame n.

Step 2 In frame n, $\{MB_0, MB_1, \dots, MB_{m_n-1}\}$ is picked out for encrypting. m_n is the number of selected MBs in frame n, defined as (4). w and h represent the width and height of the compressed video in pixels respectively. α is defined as the proportion of selection, which is the parameter in the algorithm that supports adjusting.

Step 3 Split the sequence $\{MB_0, MB_1, \dots, MB_{m-1}\}$ into several subsequences according to the NAL unit (NALU) to which the MB belongs. And then encrypt each subsequence independently with a particular key. In this way, the decryptor can do retrieval and decryption immediately after receiving a NALU, without having to wait until the GOP is fully accepted. Suppose the number of NAL units corresponding to frame n is q , the sequence consists of selected MBs will be split into subsequences $\{S_1, S_2, \dots, S_q\}$ with the length of $\{l_1, l_2, \dots, l_q\}$, satisfying the (5). And then, the MBs in subsequences are extracted from H.264 bitstream and combined into plaintexts as (6). $MB_{(s,i)}$ indicates the MB i included in the subsequence S_s . l_s is the length of S_s , which is the amount of selected MBs in NALU s . The ciphertext generated by encrypting the S_s is defined as (7). AES, as the classic symmetric encryption algorithm, is chosen to be the cryptographic algorithm, encrypting the plaintext with a 128-bit long key. For ensuring the length of the subsequences unchanged before and after encryption, it is specified not to perform padding encryption on the last block in the plaintext that less than 16 bytes in length.

Step 4 After finishing the encryption of P_s , C_s is divided into segments according to the lengths of the extracted MBs. Finally, place them back to the positions of extracted MBs in the NALU s bitstream. The SE for frame n is expressed as (8).

When the video is encoded in the CACLC mode, there is no B-frame in the GOP. Hence, the last frame will not be referenced by any frame. The BW of each block in the last frame will keep the initial value during the process of calculating. There is no doubt that the try to sort MBs in the last frame by MBW will be failure, because they have the same MBW. So, we set a constant proportion of selected MBs in the last frame to reduce the risk of information leakage, which is 0.5. Our SE method can be summarized as follows. Firstly, according to the motion reference in the reconstruction, calculate the weight of each Block and the weight of each MB in the GOP. Secondly, select MBs with high weight from all frames. Selected MBs are extracted from the slice to which they belong, then be spliced, encrypted and placed back to the bitstream. In summary, our proposed SE method for H.264 takes the MB layer as the granularity of selection and encryption. And the degree of perception of the encrypted video can be customized through adjusting the parameter of the SE algorithm appropriately.

IV. EXPERIMENTAL ANALYSIS AND RESULTS

The video content is captured either by a fixed shot or a mobile lens. In the former way, the position and focal length of the camera are constant. In the later way, not only the subject but also the camera is moving, accompanying with more motion information being encoded. Here, we adopted five CIF benchmark videos took by the later capture way to study the performance of our proposed SE algorithm, which are Mobile, Stefan, Coastguard, Basketball Pass and Foreman. All of these test sequences are encoded by H.264 baseline profile using JM18.6, with the period of IDR set to 50. In this section, the security analysis and computational complexity are detailed to verify the viability of our scheme.

A. Visual Security

Since the major drawback of SE methods is the information leakage, including motion information, object information, background information, we first verify the performance of the proposed SE scheme in visual security.



Figure 3. Visual results of the proposed SE method on the sequence Stefan with α being set to 0.05. (a) motion impact on IDR, P20, P25 and the last frame. (b) IDR, P20, P25 and the last frame of encrypted video

Fig. 3 shows the visual results of encrypted video when the parameter of the SE algorithm, α , is set to 0.05. Pictures in Fig. 3 (a) are decoding results of the video encrypted by the proposed SE. But for observing the impact on motion information, IDR frame is not encrypted. Fig. 3(b) presents decoding frames of the video which are normally encrypted in the proposed SE method. Since the moving characteristics are seriously destructed under the precise encryption of MBs that have a critical impact on motion compensation, the original moving object keeps still from the first frame to the last one. As a result, when the IDR frame is also encrypted, any information is hardly available to adversaries, as shown in Fig. 3(b). Although only 5 percent I-MBs and P-MBs of a frame are encrypted, video content is absolutely difficult to be perceived.

In Fig. 4, two benchmark sequences with intense move feature, Stefan and Basketball Pass are encrypted with α being set to 0.05, 0.04, 0.03, 0.02, and 0.01 one by one. Here, structural similarity index (SSIM) is used to measure the similarity of two videos in YUV format, original YUV video and YUV video decoded by the encrypted H.264 bitstream. The real proportion of encrypted MBs in the encoded MBs is defined as EMBR, which reflects the computational overhead. In Fig. 4(a), with the reduction of α , the impact of encryption on the video structure is gradually decreasing, which is reflected in the increase of SSIM value. Nevertheless, there is no much obvious disclosure of video content. Fig. 5(a) and (b) show the quality of videos after encryption when α is set to 0.02. Due to the amount of encrypted MBs in the IDR frame is reduced, there is a slight leakage of background information in the last frame of the GOP. But the movement trace still is hidden and the subject is hard to distinguish. Meanwhile, the EMBR declines from 6 to 3 percent (see in Fig. 4(b)). It means that the complexity is reduced exponentially.

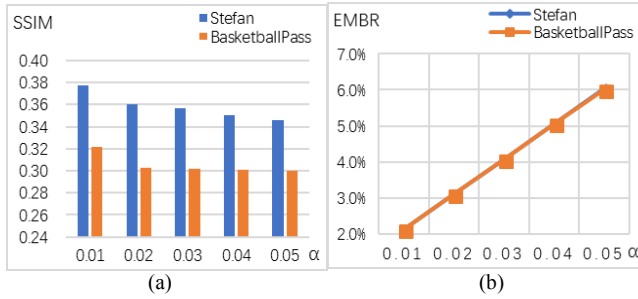


Figure 4. Variation of the SSIM and encryption ratio of the encrypted video under the different settings of α . (a) the last frames of original Basketball Pass and encrypted cipher video, with α being set to 0.02. (b) the last frames of original Stefan and encrypted cipher video, with α being set to 0.02.

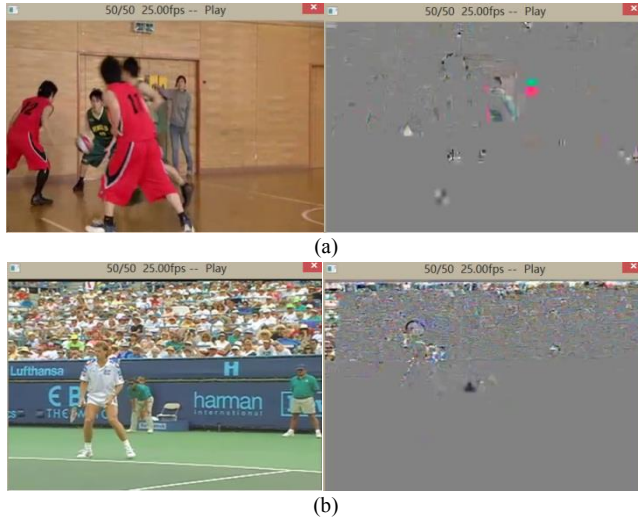


Figure 5. Visual results of the proposed SE on the sequence Stefan with α being set to 0.02. (a) the last frames of original Basketball Pass and encrypted cipher video, with α being set to 0.02. (b) the last frames of original Stefan and encrypted cipher video, with α being set to 0.02.

B. Key Space Analysis

To avoid brute force attack, the proposed SE method employs the extensive keys. First of all, each selected NALU is encrypted by AES with a 128-bit long key, so the key space for the encrypted NALU is 2^{128} . Secondly, when the frame is divided into several slices, it will be encoded into more than one NALUs for transmitting. We mark the number of NALUs corresponding to a frame is q . The key space for one encrypted frame rises to $(2^{128})^q$. At last, if the number of frames included in a GOP is n , The key space has extended to $((2^{128})^q)^n$, which is large enough to ensure the resistance regarding brute force attack.

C. Sketch Attack

The newest sketch attack against encrypted H.264 video is proposed to generate the outline of original video [3]. Authors pointed out that their sketch attack method relies on the MB bitstream size, that is, the encoding length of the MB stored in the header of each MB. In our SE scheme, the headers of the selected MBs are designed to be encrypted along with the MB data. Therefore, unless the key is compromised, our proposed encryption scheme is resistant to sketch attacks. If some video services need to attract users to subscribe higher-level video service, the video can be selectively encrypted with a lower security level. In this case, if the sketch attack is considered acceptable, the only thing that needs to be adjusted in our proposed scheme is to preserve the header of the macro block when encrypting selected MBs. And then decrypted video will be format compatible with standard decoders.

D. Computational Complexity

Since the proposed SE of H.264 video is independent of the encoder, it doesn't have any effect on the efficiency of entropy coding or compression ratio. The mainly computation overhead is in the process of encrypting data, instead of calculating BW or selecting MBs. In this section, we use EMBR and encrypted bit ratio (EDR) to measure the computational complexity of SE method. The smaller the encryption ratio is, the shorter the AES encryption takes.

Five benchmark test sequences with rich motion information and complex structure are encrypted with α being set to 0.05, the highest security level without information leakage. Among the experiment results (see Table I), the SSIM of Mobile drops most before and after encryption, the difference of which reached 0.782. It reflects that the structure of the video has been seriously damaged. Meanwhile, the EDR of Mobile is only 6.07 percent, meaning only few number of bytes are selected to be encrypted. Moreover, the EMBR changes around 6 percent because the proportion of selected MBs in each frame is set to 5% except the last frame in which is set to 50%. The difference of the numbers of MBs in the last frame resulting the slight variety of EMBR.

TABLE I. COMPARISON BETWEEN ORIGINAL VIDEO AND ENCRYPTED VIDEO

Sequence	EMBR (%)	EDR (%)	Original SSIM	Original PSNR	Encrypted SSIM	Encrypted PSNR
Mobile	5.97	6.07	0.975	35.264	0.193	12.003
Stefan	6.03	7.03	0.979	36.509	0.335	15.210
Coastguard	6.04	7.47	0.941	35.023	0.247	14.313
Basketball	6.01	8.08	0.937	38.337	0.300	15.557
Foreman	6.10	9.94	0.942	37.603	0.353	12.019

TABLE II. COMPARISON BETWEEN [12] AND PROPOSED SE METHOD

Sequence	[12]		Proposed SE	
	EMBR (%)	EDR (%)	EMBR (%)	EDR (%)
Mobile	54.0	5.30	5.97	6.07
Stefan	33.0	5.15	6.03	7.03
Foreman	42.7	8.86	6.10	9.94

TABLE III. COMPARISON BETWEEN [11] AND PROPOSED SE METHOD

Sequence	[11]		Proposed SE	
	Mobile	Foreman	Mobile	Foreman
Original PSNR	29.84	35.92	35.264	37.603
Encrypted PSNR	8.99	10.67	12.003	12.109
EDR (%)	49.640	49.175	6.07	9.94
PSNR Difference	20.850	25.250	23.261	25.584

Furthermore, we take the comparison between our proposed SE method and prior SE methods which are devised to encrypting compression videos with H.264. The motion reference struct is also used to select MBs in [12]. Due to the fixed selection, the number of selected MBs is far bigger than our proposed scheme (see Table II). Although the EDR declines by further selecting syntax elements in MBs, the MB header is exposed with more fragility in visual attack. Compared with [12], the EMBR of the proposed SE reduced by 5 to 10 times with a little rise on EDR. This also shows that the proposed SE method is more accurate on selecting MBs including key information. Another SE method related to motion vectors is presented in [11]. Due to the lack of selection strategies for motion vectors, the EDR is very high. Almost half of the bitstream are encrypted (see Table III). In contrast, our proposed SE method presents advantages in encryption efficiency. Only 6.07% and 9.94% bytes of the bitstream be encrypted for Mobile and Foreman respectively can the confidentiality be guaranteed. Furthermore, the values of PSNR difference of two test sequences in the proposed SE scheme are larger than that in [11]. This indicates that the higher visual security is achieved by the proposed SE with less computational complexity.

V. CONCLUSION

This paper proposes a novel SE method for H.264/AVC in compression domain. For a GOP, a few of MBs which are referenced more frequently than other MBs in the same frame are encrypted. Specially, security level can be adjusted by changing the proportion of selected MBs. According to the analysis of experiments results, the scheme is verified to be highly efficient in protecting confidential videos. Non-information leakage can be guaranteed by encrypting at most 10 percent bytes in a bitstream. The advantages of the proposed SE method are concluded as follows. Firstly, the proposed SE achieves the resistance to brute force attack and sketch attack and the large key space. Furthermore, it preserves the network friendliness and decoding efficiency of cipher videos by encrypting NALU with independent key. Last but not the least, the outstanding save of computation overhead is obtained through the accurate selection.

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